

CHI ZHANG

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EDUCATION

New York University | *MS in Computer Science* Sep. 2023 – Jan. 2025
Tianjin University | *BS in Construction Management* Sep. 2018 – Jun. 2023

TECHNICAL SKILLS

Languages : Python, C/C++, Java, TypeScript/JavaScript, Kotlin, Rust, SQL
Quant & ML : pandas, NumPy, PyTorch, scikit-learn, NautilusTrader, DuckDB, Prefect; backtesting, time-series, portfolio & factor-risk modeling, OpenAI API / LLM agents
Systems & Data : Kubernetes, Docker, Kafka, gRPC, Spark, Snowflake, PostgreSQL, Redis, MongoDB, AWS, GCP, CI/CD
Web : React, Next.js, Vue.js, FastAPI, Spring Boot, Node.js **Certificate:** AWS Certified Developer Associate (DVA-C02)

WORK EXPERIENCE

PayPal – Software Engineer @ San Jose Feb 2025 – Present

- Led development of the American Express transaction-fee calculation system (**Snowflake**, **PostgreSQL**) processing **\$600B+** in annual transaction volume, owning the full pipeline end-to-end from file ingestion through fee calculation to fund transfer.
- Spearheaded Braintree's net settlement project, enabling next-day fund transfers and improving cash-flow efficiency, an estimated hundreds of millions in working-capital savings.
- Owned production reliability as an on-call engineer for the \$600B+-scale fee pipeline, building a transaction-tracing and observability stack (**Sentry**, **Datadog**) and driving root-cause investigations end-to-end to prevent recurrence.
- Migrated legacy **Ruby/Kotlin** business logic to **Snowflake/Java** using an AI-native development workflow (**Claude Code** for code review, test generation, and refactoring), ensuring behavioral parity through comprehensive testing while accelerating delivery.

PayPal – Software Engineer Intern @ Chicago May 2024 – Aug 2024

- Developed the B2B backend service using **Kotlin** and **Spring Boot** to calculate and deduct transaction fees and generate compliance reports for PayPal's merchants who use our billing gateway.
- Built an efficient **Kafka** message queue for accurate transaction-event processing, using Kafka proxies within **Kubernetes** for distributed node operations and stability.
- Optimized logic of transaction verification, settlement, chargeback, and audit compliance with newly designed multiple **Airflow** DAGs for efficient scheduling of **Spark** jobs.

PROJECTS

Quant-Stroller – Multi-Market Quant Research Platform 🌐 | *Python, NautilusTrader, DuckDB, React* Jun 2026 – Present

- Architected a quantitative research and backtesting platform (Python, 66K+ lines, 470+ test files) spanning US equities, crypto, China A-shares, and FX; narrow DataSource / Factor / Strategy / Broker seams onboard a new market or vendor as a thin adapter with zero strategy-code changes.
- Enforced anti-overfitting discipline via a dual-gate pipeline (purged cross-validation, Deflated Sharpe Ratio, PBO, plus execution replay on **NautilusTrader**), logging 340+ experiments to an append-only ledger with calibrated per-market transaction-cost models.
- Validated a China small-cap monthly strategy through both gates (excess Sharpe 1.39 net of 10 bps costs); deployed to a live paper-trading loop with **Alpaca/CCXT** broker routing, pre-trade risk checks, and 5-factor risk reporting.
- Built a point-in-time data plane (raw → bars → panel) on **DuckDB**/Parquet with a 2,300+ factor catalog across 9 vendors; ran 5,500-stock A-share panels and **PyTorch** training on an 8-GPU server with self-hosted CI; shipped a **FastAPI** + **React** research console.

AikiCard – AI Learning Platform 🌐 | *React, Next.js, TypeScript, OpenAI API, Vercel* Jun 2025

- Built a web platform using **React** and **Next.js** with **TypeScript** that transforms EPUB books into interactive learning content, serving users with fast global access via **Vercel** edge caching.
- Implemented a spaced-repetition algorithm (SM-2) with progress tracking and adaptive review scheduling.
- Designed a responsive UI with **React** and **TailwindCSS**, optimized for performance across devices.

Aiki – AI Driven Memo Card 📱 | *iOS, SwiftUI, LLM, SwiftData* Mar 2024

- Developed an iOS flashcard app using **OpenAI's o3** for intelligent content generation and adaptive learning paths.
- Implemented local storage with **Core Data** and **CloudKit** sync, and an intuitive **SwiftUI** UI with **StoreKit** in-app purchases, achieving a 4.5+ App Store rating.